Saahil Sanganeriya

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Education

Georgia Institute of Technology — *Atlanta, GA*
Master of Science in Computer Science (ML) GPA 4.00 — *Dec 2025*

Georgia Institute of Technology — *Atlanta, GA*
Bachelor of Science in Computer Science (ML, Theory) GPA 3.76 — *Dec 2024*

Experience

Software Engineer, Applied AI — *San Francisco, CA*
Mercor — *Mar 2026 – present*
Team: Data Acquisition, delivering anonymized multi-terabyte enterprise data (billions of tokens) to a frontier AI lab for post-training

- Designed and built entity resolution, which links every name, email and handle to the real person behind it (LLM type classification, FAISS embedding search, Leiden graph clustering), plus its eval suite; raised accuracy on the labeled eval set from **85% to 95%**
- Trained LightGBM and PyTorch match scorers on a 257-feature Rust library, served via ONNX and run in production shadow mode
- Lifted B-cubed F1 from **0.34 to 0.84** by making matching blockwise: MinHash-LSH groups likely matches, one LLM call clusters each block, and merges need agreeing votes (a second mode uses correlation clustering) so one bad verdict can't join two people
- Rebuilt it as a bounded-memory Go service that scales horizontally on Temporal across GKE and EKS; resolved **88M mentions** in one run and halved wall time (**4h13m to 2h11m**) by tuning its LLM judges
- Cut LLM provider failures **22x** (24.6 to 1.1/min, same throughput) over **~3M LLM calls** per run with an adaptive (AIMD) concurrency controller across providers, plus versioned Temporal workers and a custom rollout controller for safe mid-run deploys
- Built the fake-identity generator, giving each person one consistent persona through format-preserving generators for 90+ PII types and length-preserving LLM-written names and addresses; expert reviewers kept **95%** of its replacements on 675 documents
- Created a synthetic-data engine in which an LLM agent designs a fictional company and writes its Slack, Gmail and Linear history while code places every PII value, so labels are exact by construction; its **1.6M labeled mentions** became the team's benchmark
- Redesigned code anonymization to rewrite git histories (tree-sitter, Aho-Corasick) and hand each repo to a sandboxed LLM coding agent that runs its tests; running **~200 repos** at once on ECS, it cut a 55k-commit repo from **3+ hours to 19 min**

Data Engineer Intern — *Seattle, WA*
Amazon — *May 2025 – Aug 2025*

- Engineered a real-time streaming pipeline (Apache Flink in Java, Kinesis, Lambda) built to replace an hourly batch job for Amazon's workforce management, processing cross-region SNS events at **600–900ms** latency against a 120s SLA
- Chose Flink over Firehose, OpenSearch Ingestion, Data Prepper and direct Lambda on latency, exactly-once guarantees and cost
- Built 5 custom OpenSearch sinks with AWS IAM SigV4 auth and Painless scripts that reject out-of-order events, for exactly-once results with automatic failover and Lambda headroom for ~300K events/min
- Designed a fault-tolerant dual-path architecture (real-time + S3 backup) spanning AWS regions (Australia to US) with retries, CloudWatch alerting, live OpenSearch dashboards and dead-letter queue replay

Data Engineer Co-op — *Atlanta, GA*
The Coca-Cola Company — *May 2023 – Aug 2024*

- Built a Haystack chatbot that answers employees' questions from internal docs (BM25 over Elasticsearch, RoBERTa reader)
- Architected Django REST APIs over 300+ global supply-chain datasets, replacing manual data requests with BI self-service
- Cut support tickets **63%** and ETL runtime from **9 to 6.5 hours** by rebuilding a Python CRUD library and its monitoring platform
- Built Azure Data Factory ETL pipelines that process 300+ datasets daily with PySpark and SQL on Databricks, cutting the transformation step's runtime **60%**, with CI/CD and automated data validation

Research and Publications

Graduate Researcher — *Atlanta, GA*
Georgia Institute of Technology — *Aug 2024 – Dec 2025*
VandeLoo AD*, Malta NJ*, **Sanganeriya S**, Aponte E, van Zyl C, Xu D, Forest C. *SAMCell: Generalized label-free biological cell segmentation with segment anything.* PLOS ONE, 2025

- Developed SAMCell: proposed its distance-map regression with watershed post-processing and trained it, beating Cellpose and StarDist on LIVECell (**0.77 vs 0.68** OP-CSB) and nearly doubling Cellpose's zero-shot score on an unseen microscope (**0.60 vs 0.32**)
- Fine-tuned SAM's ViT image encoder and mask decoder with a frozen prompt encoder, removing manual prompting for hundreds of cells per image; trained on 5,000+ LIVECell images in PyTorch with CLAHE preprocessing and 256x256 sliding-window inference
- Created two cross-microscope benchmark datasets and an open-source Napari plugin that biologists run without ML expertise

Projects

swe-sim: Graded Coding-Agent Tasks from Real GitHub Issues — *Sept 2026*

- Built a pipeline that turns underspecified GitHub issues into coding-agent tasks, with an LLM stakeholder the agent can question
- Graded each patch offline, dropping edits to protected tests and paying out only when four records of the hidden-test run agree
- Ran it on Temporal, Modal and Cloud Run; the first hosted task (29 trials, $3.01) exposed a task defect that a new gate now rejects

travel-agent-gym: An RL Environment for a Travel Agent — *Aug 2026*

- Built a Gymnasium environment where an agent interviews a client with hidden needs, books a trip, then rides out its delays
- Made illegal bookings impossible (action masks, integer-cent ledger) and exposed the action set to LLM agents as tool calls

Reinforcement Learning Skill Discovery for Atari Games — *Jan 2025 – May 2025*

- Adapted METRA skill discovery to discrete-action Atari with SAC-Discrete, to learn distinct skills without hand-designed rewards
- Diagnosed representation collapse with t-SNE and silhouette analysis, and traced it to a weak pixel encoder by swapping in ResNet-18

Skills

Languages & Tools: Python, Go, Rust, Java, SQL, C/C++, JavaScript, Scala; Git, Linux, Datadog, CloudWatch, Airflow, CI/CD
ML & LLMs: PyTorch, LightGBM, ONNX, Hugging Face Transformers, FAISS, scikit-learn, LLM APIs (Claude, Gemini, Bedrock)
Distributed Systems: Temporal, Kubernetes (EKS, GKE), Helm, KEDA, Karpenter, Terraform, Docker, Modal, Spark, Flink, Kinesis
Cloud & Data: AWS (ECS, S3, Lambda), Azure (ADF, ADLS), Databricks, Parquet, DuckDB, PostgreSQL, OpenSearch, Snowflake